Huawei Access Node QoS/Congestion Event Troubleshooting and Resolution SOP for Hebbal Edge Site

Node: HEB-ACC-40 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9777

Description

The occurrence of QoS/Congestion events, such as microbursts, interface tail drops, and QoS queue starvation, on the Huawei Access Node (HEB-ACC-40) at Hebbal Edge, is precipitated by a complex interplay of factors encompassing both the hardware and software realms. This failure domain is particularly pertinent in the context of access roles, where the aggregation of subscriber traffic from diverse sources converges, thereby amplifying the potential for congestion. The Huawei equipment, leveraging its proprietary CLI dialect, necessitates a nuanced understanding of its command structure to effectively diagnose and mitigate these issues. The root cause of such failures can often be attributed to misconfigurations in QoS policies, inadequate interface bandwidth allocation, or unforeseen spikes in traffic patterns. Given the minor severity of the alarm (ALARM_9777), a methodical approach to troubleshooting and resolution is paramount to prevent escalation and ensure service continuity.

Pre-requisite Checks

- Verify the physical integrity and connectivity of all interfaces on the affected node.
- Confirm the operational status of redundant links to ensure no single point of failure.
- Dispatch remote hands to the Hebbal Edge site for potential hardware interventions.
- Review recent configuration changes or software updates that may have precipitated the issue.

Potential Root Causes

Cause	Probability
Misconfigured QoS Policies	High
Inadequate Interface Bandwidth Allocation	Medium
Unforeseen Traffic Pattern Changes	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos interface, Expected Output: QoS configuration and statistics for each interface., Description: This command isolates the QoS configuration applied to each interface, allowing for the identification of potential misconfigurations or bottlenecks in the QoS policies.
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Command: display interface brief, Expected Output: Brief overview of all interface statuses and traffic statistics., Description: Provides a snapshot of interface health and utilization, which can indicate interfaces experiencing congestion or errors.

Command: display traffic-statistics interface, Expected Output: Detailed traffic statistics for each interface., Description: Offers insights into traffic patterns and volumes, aiding in the identification of interfaces that may be experiencing microbursts or other forms of congestion.
```

Resolution Steps

Step 1: Step: Adjust QoS Policies, Command: qos policy {policy-name} class {class-name} shaping {shaping-rate}, Description: Adjust the QoS policies to ensure adequate bandwidth allocation and proper traffic shaping to prevent congestion. This involves modifying existing policies or creating new ones as necessary to align with the current traffic demands.

Step 2: Step: Upgrade Interface Bandwidth, Command: interface {interface-name} bandwidth {bandwidth-value}, Description: Increase the bandwidth allocation of interfaces identified as bottlenecks. This may involve physical upgrades or reconfiguration of existing links to improve capacity.

Step 3: Step: Implement Traffic Management, Command: traffic-policy {policy-name} inbound, Description: Apply traffic management policies to interfaces to manage and prioritize traffic flows effectively, preventing congestion and ensuring critical traffic is prioritized.

Post-Check Verification

```
Command: display qos interface, Expected Output: Updated QoS configuration reflecting changes made., Description: Verifies that QoS policies have been correctly adjusted and are operational.

Command: display interface brief, Expected Output: Improved interface statuses and reduced error rates., Description: Confirms that interface health has improved following the implementation of resolution steps.

Command: display traffic-statistics interface, Expected Output: Traffic statistics indicating reduced congestion., Description: Validates that the implemented changes have effectively mitigated congestion and improved overall network performance.
```

Roll-back Procedures

- Step: Revert QoS Policies, Command: qos policy {original-policy-name} class {original-class-name} shaping {original-shaping-rate}, Description: Revert QoS policies to their original configuration if changes do not yield the desired outcome, to prevent further disruption.

- Step: Revert Interface Bandwidth, Command: interface {interface-name} bandwidth {original-bandwidth-value}, Description: Restore the original bandwidth configuration of interfaces that were modified, ensuring that any changes made are fully reversible.

- Step: Remove Traffic Management, Command: undo traffic-policy {policy-name} inbound, Description: Remove any newly implemented traffic management policies if they are deemed ineffective or cause unforeseen issues, reverting to the previous state.